References

[1] CJL Murray, KS Ikuta, F Sharara, L Swetschinski, GR Aguilar, et al. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *The Lancet*, 399(10325):629–655, 2022. https://doi.org/10.1016/S0140-6736(21)02724-0.

[2] M Naghavi, SE Vollset, KS Ikuta, LR Swetschinski, AP Gray, et al. Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. *The Lancet*, 404(10459):1199–1226, 2024. https://doi.org/10.1016/S0140-6736(24)01867-1.

[3] KE Arnold, G Laing, BJ McMahon, S Fanning, DJ Stekel, et al. The need for one health systems-thinking approaches to understand multiscale dissemination of antimicrobial resistance. *The Lancet Planetary Health*, 8(2):e124–e133, 2024. https://doi.org/10.1016/S2542-5196(23)00278-4.

[4] MJ Shepherd, T Fu, NE Harrington, A Kottara, K Cagney, et al. Ecological and evolutionary mechanisms driving within-patient emergence of antimicrobial resistance. *Nature Reviews Microbiology*, 22(10):650–665, 2024. https://doi.org/10.1038/s41579-024-01041-1.

[5] IH Spicknall, B Foxman, CF Marrs, and JNS Eisenberg. A modeling framework for the evolution and spread of antibiotic resistance: literature review and model categorization. *American Journal of Epidemiology*, 178(4):508–520, 2013. https://doi.org/10.1093/aje/kwt017.

[6] EI Nielsen and LE Friberg. Pharmacokinetic-pharmacodynamic modeling of antibacterial drugs. *Pharmacological Reviews*, 65(3):1053–1090, 2013. https://doi.org/10.1124/pr.111.005769.

[7] C Witzany, J Rolff, RR Regoes, and C Igler. The pharmacokinetic–pharmacodynamic modelling framework as a tool to predict drug resistance evolution. *Microbiology*, 169(7):001368, 2023. https://doi.org/10.1099/mic.0.001368.

[8] A Handel and P Rohani. Crossing the scale from within-host infection dynamics to between-host transmission fitness: a discussion of current assumptions and knowledge. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1675):20140302, 2015. https://doi.org/10.1098/rstb.2014.0302.

[9] N Mideo, S Alizon, and T Day. Linking within- and between-host dynamics in the evolutionary epidemiology of infectious diseases. *Trends in Ecology & Evolution*, 23(9):511–517, 2008. https://doi.org/10.1016/j.tree.2008.05.009.

[10] HK Alexander and RC MacLean. Stochastic bacterial population dynamics restrict the establishment of antibiotic resistance from single cells. *Proceedings of the National Academy of Sciences*, 117(32):19455–19464, 2020. https://doi.org/10.1073/pnas.1919672117.

[11] P Czuppon, T Day, F Débarre, and F Blanquart. A stochastic analysis of the interplay between antibiotic dose, mode of action, and bacterial competition in the evolution of antibiotic resistance. *PLoS Computational Biology*, 19(8):e1011364, 2023. https://doi.org/10.1371/journal.pcbi.1011364.

[12] P Macheras and AA Tsekouras. *Revising Oral Pharmacokinetics, Bioavailability and Bioequivalence Based on the Finite Absorption Time Concept*. Springer, Cham, 2023.

[13] P Chryssafidis, AA Tsekouras, and P Macheras. Re-writing oral pharmacokinetics using physiologically based finite time pharmacokinetic (pbftpk) models. *Pharmaceutical Research*, 39(4):691–701, 2022. https://doi.org/10.1007/s11095-022-03230-0.

[14] K Drlica. The mutant selection window and antimicrobial resistance. *Journal of Antimicrobial Chemotherapy*, 52(1):11–17, 2003. https://doi.org/10.1093/jac/dkg269.

[15] DT Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977. https://doi.org/10.1021/j100540a008.

[16] LJS Allen. *An Introduction to Stochastic Processes with Applications to Biology*. CRC Press, Boca Raton, FL, 2 edition, 2010.

[17] World Health Organization. *Adherence to Long-Term Therapies: Evidence for Action*. World Health Organization, Geneva, 2003.

[18] JA Gilsoul, HR Tung, and SD Lawley. Mitigating antibiotic resistance from missed doses. *Journal of Pharmacokinetics and Pharmacodynamics*, 53(3):21, 2026. https://doi.org/10.1007/s10928-026-10027-w.